



Metal vs Plastic

It's like trying to distract a dog from the bacon with yesterday's chicken. iPhone 5c just can't seem to compete. <http://dgit.in/1cwzELz>

War of the tablets

Read further to see the comparison between the new iPad Air, Nexus 7 and Surface 2 series: <http://dgit.in/1bdUn59>

Smart SoHo

How it does that is by having read and write caches like every other system but it continually indexes each file and creates checksums which are stored away from the actual file. So in cases of passive data corruption which occurs randomly even when the file is not being accessed, the file system can detect and heal the corrupt data. Also you don't need hardware RAID, infact it is recommended that you don't set it up at all since ZFS has it's own software RAID functionality called RAID-Z1 and RAID-Z2 which has overcome the drawbacks of RAID 5.

It also features deduplication and compression! Deduplication is where the system first creates a checksum for a new file to be written and then checks if the file has been written before. When there is a previous copy found the new one is not written and just an entry for the file location is made. Which is why you shouldn't backup the NAS data onto the same NAS device because the deduplication won't allow a new copy to be written. However, deduplication being the memory intensive operation that it requires a lot of memory (5GB RAM per TB of data!). So most people prefer to turn it off and then backup data onto the same NAS device till it can be moved elsewhere.

Hardware needed

The expenditure on the hardware comes down to what all features you wish to enable on your FreeNAS installation. Installing a 32-bit OS limits the capabilities severely since it can only address 4GB of RAM which means after keeping aside 1 GB for the OS there is 3GB of RAM left which can manage 3TB of storage space, in case you wish to use a 32-bit installation do stick to a UFS file system. Now we'll look at the 64-bit installation. ZFS can be used properly on a 64-bit installation and requires a minimum of 8GB of RAM. ZFS makes use of all the RAM you throw at it so the more it gets the better it can perform. If you wish to enable deduplication on ZFS then you need to get 5GB of RAM per TB of storage space which amounts to an awful lot once you go beyond 5TB. Also you might have to spend a little extra on a newer motherboard which can handle a lot more RAM. As for speed – 1600MHz is fine.

The processor should be rated at 1.6 GHz minimum for normal operation, the more features you enable the more processing power the device needs, going all the way to 3.2 GHz at peak loads is normal for deduplication. Intel or AMD doesn't matter since all the processor is doing is hashing files and nothing else.



The WD Red Drive

Hard drive manufacturers like Western Digital, Seagate, Toshiba have come up with hard drives specifically suited for NAS drives. The WD Red drive is one such device which is suited for a NAS implementation. It comes in multiple capacities ranging from 1TB to 4TB, the RPM is managed intelligently given it's IntelliPower technology which ramps up RPM when data is to be accessed and reduces RPM while idling. Also it features 3D Active Balance technology which leads to reduced wobbling. This not only saves power but also reduces the overall temperature and vibration inside the NAS device – both being harmful to hard drives in particular.

Better yet, these drives are tested and certified by all major third-party NAS manufacturers. So you can use them without any worry. They are better suited for 1-5 bay NAS implementations, in case you wish to add more drives than 5 you might want to consider enterprise solutions like WD Se and WD Re.

Set it up!

1. Download

Obtain whichever OS you want, we are using FreeNAS here. Get the ISO or the flash drive image. Use 7-zip to extract the

.xz file. Then use Win32DiskImager to use the uncompressed .img file and create a bootable USB flash drive.

2. Install all hardware

Get your computer configuration together, ensure that there is enough RAM as per your storage needs. Plug in the Ethernet cable and ensure all the WD Red hard drives are connected and are mounted securely.

3. Install FreeNAS

Boot up and change boot priority (usually by pressing F11) to boot from the flash drive. There will be a lot of random text flowing across the screen till you get to the ASCII art that spells out FreeNAS. Press 1 and then Enter to begin installation or FreeNAS will auto install, anyway.

4. Configure

Within a few minutes, FreeNAS will have installed and you'll be shown a console screen with an IP address mentioned at the bottom. Note this address down and try to access the NAS device from another computer's web browser. If you can access it then unplug your monitor and keyboard from the NAS device. You are done here.

5. Web interface

The default user id is "admin" and the password is "freenas". From this web management console you can get everything done. You can create partitions, create backups, schedule regular disk image creations or by means of various plugins add more features. Change the default id and password for security reasons.

6. Plugins

There are a lot of plugins available which should enable you to use your NAS as a media server or a web server. Install Plex to enable a media server GUI tool on the web interface or you could install Lamp and get a fully functional web server for your hosting needs.

Conclusion

That's all there is to it! It's not a big tutorial because there simply isn't enough steps involved, so get cracking and build your own NAS device for cheap! 